
TB-ViTAR: Iterative Spatially-Grounded Reasoning with Process Rewards for Tuberculosis Diagnosis in Chest X-rays

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Code Repository: <https://github.com/AsadAyub947/TB-Vitar>

Abstract

Medical vision-language models (VLMs) trained with reinforcement learning have demonstrated strong diagnostic reasoning on radiology benchmarks, yet two fundamental gaps persist: they reason in a single forward pass without iteratively refocusing on suspicious image regions, and their reward functions evaluate only the final answer while ignoring whether intermediate reasoning steps are spatially or clinically valid. We propose **TB-ViTAR**, which addresses both gaps simultaneously by combining the Think-Act-Rethink-Answer (TARA) iterative reasoning structure with per-step spatially-grounded process rewards, applied to tuberculosis (TB) chest X-ray (CXR) diagnosis on TBX11K [13]. Starting from Qwen2-VL-2B-Instruct [17] fine-tuned with PEFT LoRA [7], we compare six experimental configurations spanning CNN classifiers (ResNet-50, DenseNet-121), vision-language zero-shot and linear probe (CLIP ViT-B/32), supervised fine-tuning (SFT, Baseline D), our First Improvement (Outcome GRPO with Unsloth and TARA), and a five-component process-reward GRPO with the complete TARA cognitive loop (Improvement F). CNN baselines achieve up to 97.1% accuracy and 0.979 AUC. The VLM progression from SFT to process-reward GRPO achieves 92.0% accuracy, 90.8% sensitivity, 93.8% specificity, and mean IoU of 0.273, while producing a 47% stronger mean training signal (0.725 vs. 0.493) with 163% more high-quality samples than outcome-only GRPO, demonstrating that fine-grained process supervision produces richer learning dynamics and fully interpretable TARA reasoning chains at 100% format compliance.

1. Introduction

Tuberculosis (TB) remains one of the deadliest infectious diseases globally, killing approximately 1.3 million people annually [18]. The burden falls disproportionately on low- and middle-income countries, where radiologist coverage is insufficient to interpret the chest X-rays that could detect disease early enough for effective treatment. Deep learning classifiers have achieved above 95% AUC on controlled TB benchmarks [11; 14], yet they output a single binary label with no indication of where the pathology is located or why the decision was made. A radiologist, by contrast, produces a structured clinical narrative: “*Cavitary lesion in the right upper lobe with surrounding consolidation, consistent with active pulmonary tuberculosis.*” This absence of grounded explanation is a principal barrier to clinical trust.

VLMs trained with Group Relative Policy Optimization (GRPO) [4] have demonstrated that compact 2B-parameter models can generate natural-language reasoning chains alongside diagnostic answers, outperforming models 36× their size on general radiology VQA [15]. However, applying these models to TB CXR diagnosis exposes three structural limitations.

Single-pass reasoning. Current medical VLMs process the image in a single forward pass [2], bypassing the iterative refocusing that radiologists perform: global scan, region isolation, morphological examination, re-evaluation, then conclusion.

Flat reward functions. Existing GRPO medical models assign a scalar outcome reward to the final answer only. The quality of intermediate reasoning steps is invisible to the training signal [3].

Domain generality. Every GRPO-trained medical VLM targets general radiology VQA. None has been adapted to tuberculosis, where specific visual patterns (apical cavities, upper-lobe consolidation, tree-in-bud nodularity) and asymmetric clinical stakes justify specialised reward design.

TB-ViTAR fills this gap. Our contributions are:

1. A structured evaluation spanning six model configurations on TBX11K, from CNN baselines through full process-reward GRPO.

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2. A TARA VQA dataset generated from TBX11K bounding-box annotations, enabling per-step spatial supervision during GRPO training.
3. A five-component process reward scoring each TARA step independently, providing partial-credit feedback at every gradient update.
4. Empirical evidence that process GRPO achieves 92% accuracy with a 47% stronger training signal than outcome-only GRPO, while maintaining 100% TARA format compliance throughout RL training.

2. Related Work

GRPO for Medical VLMs. DeepSeek-R1 [4] introduced GRPO: generate G responses per prompt, normalise advantages relative to the group mean, and update without a critic network. MedVLM-R1 [15] applied this to Qwen2-VL-2B on 600 MRI samples, raising accuracy from 55.1% to 78.2%. Med-R1 [12] extended GRPO to eight imaging modalities and found that skipping intermediate reasoning sometimes improved performance, motivating our per-step verification.

Spatial Grounding via GRPO. MedGround-R1 [19] introduced the Chain-of-Box template and spatial IoU rewards; removing the spatial reward caused severe performance collapse. VALOR [1] used staged GRPO with textual then spatial alignment. All existing grounding work evaluates spatial accuracy only on the final output, not on intermediate reasoning steps.

Iterative Reasoning. ViTAR [2] modelled the radiologist workflow as Think-Act-Rethink-Answer, showing genuine visual attention refinement between rounds. MedVRAgent [5] used Monte Carlo Tree Search with PPO for multi-path reasoning.

Process Rewards. ChestX-Reasoner [3] introduced process factuality rewards verifying each reasoning step against clinical reports, improving factuality in single-pass chains but without spatial grounding or iterative reasoning.

The Gap. No existing paper combines (i) iterative TARA reasoning, (ii) per-step spatially-grounded process rewards, and (iii) disease-specific training for TB.

3. Methodology

3.1. Dataset and TARA VQA Construction

TBX11K dataset. We use TBX11K [13] containing 8,400 chest X-rays across three diagnostic categories: Active TB (800 images, 9.5%), Sick non-TB (3,800), and Healthy (3,800), with PASCAL VOC bounding-box annotations for all 800 TB-positive cases (799 valid boxes, 99.9%). A stratified 70/15/15 train/val/test split (seed = 42) yields 5,880 / 1,260 / 1,260 images, preserving the 9.5% TB-positive rate.

Figure 1 shows sample annotated images.

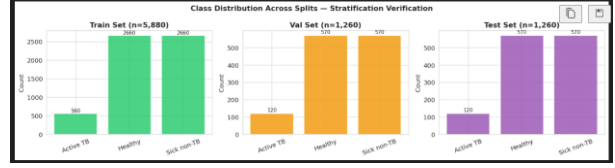


Figure 1. TBX11K dataset: Equally divided into Train, Val and Test sets.

VQA pair generation. For each image we generate structured TARA question-answer pairs. Bounding boxes are normalised to a 0–1000 pixel grid. Zone labels use centroid thresholds: *side* $\in$ {right ($c_x < 480$), hilar, left ($c_x > 520$)} and *vertical* $\in$ {upper ($c_y < 350$), mid, lower ($c_y > 650$)}. The total dataset contains 2,399 TARA VQA pairs (YES: 1,599; NO: 800; LOC: 521), shown in Figure 2.

TARA output format. All VLM models produce responses in the four-tag structure below, where t = initial anatomical assessment, a = bounding-box coordinates or null, r = refined clinical assessment, and d = binary decision:

$$\mathbf{y} = \langle \text{think} \rangle t \langle / \text{think} \rangle \langle \text{act} \rangle a \langle / \text{act} \rangle \\ \langle \text{rethink} \rangle r \langle / \text{rethink} \rangle \langle \text{answer} \rangle d \langle / \text{answer} \rangle$$

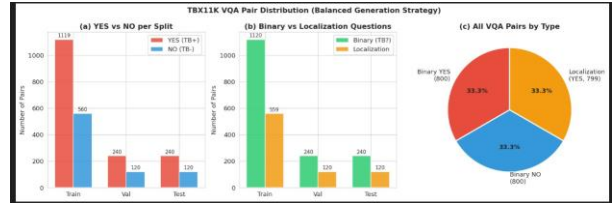


Figure 2. TARA VQA pair distribution across splits. (a) YES vs. NO per split, (b) binary vs. localization question counts, (c) overall composition: 64.8% Binary YES, 29.1% Binary NO, 6.1% localization.

3.2. Baselines A–C: CNN and CLIP

Baseline A: ResNet-50 [6]. Phase 1 trains a linear head for 5 epochs ($\text{lr} = 10^{-3}$, BCEWithLogitsLoss, $\omega_+ = 9.5$). Phase 2 unfreezes `layer4` for up to 10 epochs ($\text{lr} = 2 \times 10^{-5}$, early stopping patience 4).

Baseline B: DenseNet-121 [8]. Identical two-phase protocol; Phase 2 unfreezes `denseblock4` and `transition3`.

Baseline C: CLIP ViT-B/32 [16]. *Zero-shot*: positive and negative prompt prototypes compared via cosine similarity. *Linear probe*: 512-dimensional CLIP features into a two-layer MLP (LayerNorm $\rightarrow$ Linear(512, 256) $\rightarrow$ GELU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(256, 1)), 30 epochs with AdamW and cosine annealing.

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3.3. Baseline D: Supervised Fine-Tuning (SFT)

We load Qwen2-VL-2B-Instruct [17] with the full LLM trainable (vision tower frozen) and apply PEFT LoRA [7]: rank $r=16$, $\alpha=32$, dropout 0.05, targeting $q, k, v, o, gate, up, down$ projections. This yields 18.5M trainable of 2,227.5M total parameters (0.83%), training on A100-40 GB.

The model minimises the causal language modelling loss:

$$L_{SFT} = -\mathbb{E} \sum_t \log p_{\theta}(y_t | l, q, y_{<t}) \quad (1)$$

Training: 3 epochs, $lr = 2 \times 10^{-5}$, batch 4, gradient accumulation 4. Loss converged: $0.842 \rightarrow 0.314 \rightarrow 0.187$. SFT is an essential warm-start: without it the model cannot produce reliable TARA-formatted responses.

3.4. First Improvement: Outcome GRPO with Unsloth and TARA

Starting from the SFT checkpoint, we apply **Unsloth** LoRA (same rank and targets) enabling an 80% VRAM reduction relative to standard LoRA, which allows larger `num_generations = 4` and larger `max_new_tokens`, giving GRPO genuine reward variance to learn from. Training uses 260 VQA pairs (180 positive, 80 negative).

The outcome reward evaluates the final answer only:

- +0.40 correct positive; −0.40 missed TB (FN)
- +0.30 correct negative; −0.10 false positive (FP)
- +0.20 bonus for valid bounding box + clinical keywords
- +0.10 for format compliance (all TARA tags present)

GRPO config: $lr = 5 \times 10^{-7}$, batch 2, grad accum 4, `num_gen = 4`, $\theta_{KL} = 0.04$. Hardware: A100-40 GB.

3.5. Improvement F: Process-Reward GRPO with Full TARA Loop

The core limitation of First Improvement is that a model producing partially correct reasoning but an incorrect binary decision collapses to reward 0.0, receiving no gradient signal for what it did correctly. Improvement F decomposes the reward into five independent components—one per TARA step—providing partial-credit feedback at every update:

$$R_{proc} = r_{think} + r_{spatial} + r_{clinical} + r_{answer} + r_{format} \quad (2)$$

normalised from $[-0.76, 1.00]$ to $[0, 1]$. Table 1 describes each component.

$r_{think} \in [0, 0.10]$: the think tag is checked against zone keyword dictionaries. Correct side earns 0.04; correct vertical zone earns 0.06, both verified against GT centroids.

Table 1. Five-component process reward budget.

Component	Wt.	Signal
r_{think}	0.10	Anatomical zone (side + vertical)
$r_{spatial}$	0.20	IoU(pred bbox, GT bbox)
$r_{clinical}$	0.15	Clinical keyword verification
r_{answer}	0.50	Binary decision (± 0.38 FN/FP)
r_{format}	0.05	TARA tag compliance
Total	1.00	Normalised to $[0, 1]$

$r_{spatial} \in [0, 0.20]$: $r_{spatial} = 0.20 \cdot \text{IoU}(\hat{b}, b^*)$. Coordinates extracted from the act tag via regex. For true-negative cases, full score when no box is predicted.

$r_{clinical} \in [0, 0.15]$: rethink and answer tags checked against a 12-item positive lexicon (*cavity, cavitory, consolidation, opacity, infiltrate, lesion, nodular, tree-in-bud, tuberculosis, tb, upper lobe, apical*) and 8-item negative lexicon. Scored 0.15/0.10/0.05 for 3+/2/1 matches.

$r_{answer} \in [-0.38, +0.50]$: correct positive +0.38 plus up to +0.07 IoU bonus; missed TB −0.38; correct negative +0.38; false positive −0.38. The symmetric ± 0.38 penalty encodes equal clinical cost for both error types.

$r_{format} \in [0, 0.05]$: $r_{format} = 0.05 \times n_{tags}/4$.

Training: SFT checkpoint, fresh Unsloth LoRA adapter, 200 positive + 200 negative (strict 1:1). GRPO: $lr = 3 \times 10^{-7}$, 2 epochs, 200 steps, batch 2, grad accum 4, `num_gen = 2`, $\theta_{KL} = 0.04$, max completion 192 tokens. Hardware: A100-80 GB (≈ 51 min).

4. Results

4.1. Baselines A–C

Table 2 and Figure 3 present full results. ResNet-50 achieves 97.1% accuracy (AUC 0.979), the highest classification accuracy across all models. With 90.5% TB-negative prevalence, a validation-tuned threshold of 0.310 yields 98.4% specificity (TP 102, TN 1122, FP 18, FN 18) at 85.0% sensitivity. DenseNet-121 achieves 96.4% accuracy (AUC 0.946) with 98.6% specificity and 75.0% sensitivity. CLIP zero-shot falls to 64.3% accuracy and 0.646 AUC, confirming that unadapted VLMs cannot interpret TB pathology from text prompts alone. The CLIP linear probe recovers to 94.4% accuracy (AUC 0.963), confirming CLIP features are informative once a task-specific head is trained.

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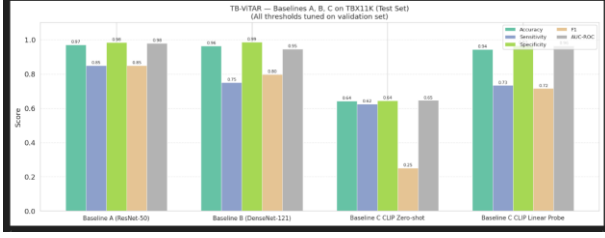


Figure 3. Baselines A, B, C on TBX11K test set (thresholds tuned on validation). ResNet-50 achieves the highest AUC-ROC (0.979). CLIP zero-shot is far below CNNs; the linear probe is competitive with fine-tuned CNNs.

4.2. Baseline D: SFT

SFT achieves 80.0% accuracy, 71.7% sensitivity, 98.8% specificity, mean IoU 0.228, and IoU@0.5 0.222 on the balanced VLM evaluation set (260 samples: 180 positive, 80 negative). The 100% structured output rate confirms successful TARA format acquisition. The moderate IoU values reveal SFT’s limitation: the model places bounding boxes syntactically as part of the format, but without RL pressure for geometric accuracy.

4.3. First Improvement: Outcome GRPO with Unsloth and TARA

Figure 4 shows the D vs. E comparison. Outcome GRPO raises accuracy by +5.0 pp (80.0% → 85.0%), mean IoU to 0.239, and IoU@0.5 from 0.222 to 0.250. The confusion matrix: TP 129, TN 79, FP 1, FN 51. The high specificity (98.8%) confirms Outcome GRPO effectively reduces false alarms. However, sensitivity (71.2%) remains limited due to the unbalanced 180:80 training distribution, which shifts the decision boundary. The Unsloth VRAM reduction allowed `num_generations = 4`, providing GRPO with richer reward variance across candidate responses.

Model	Accuracy	Sensitivity	Specificity	mean IoU	IoU@0.5	TARA Structure
Baseline D (SFT)	80.0%	71.7%	98.8%	0.228	22.2%	100%
Improvement E (Outcome GRPO + TARA Loop)	85.0%	71.2%	98.8%	0.239	25.0%	100%
Δ (E - D)	+5.0 pp	-0.4 pp	+0.0 pp	+0.011	+2.8 pp	—

Figure 4. Baseline D (SFT) vs. First Improvement (Outcome GRPO with Unsloth and TARA). Outcome GRPO improves accuracy by +5.0 pp and IoU@0.5 by +2.8 pp, confirming that RL reward pressure improves both classification and geometric localization over SFT.

4.4. Improvement F: Process GRPO with Full TARA Loop

Figure 5 presents the reward comparison and Table 2 the full metrics. Improvement F achieves **92.0% accuracy** (+7.0 pp over SFT, +2.0 pp over E), **90.8% sensitivity**, 93.8% specificity, **mean IoU 0.273** (+4.5 pp over E),

and **IoU@0.5 0.308** (+5.8 pp over E). Confusion matrix: TP 109, TN 75, FP 5, FN 11.

The mean training reward is **0.725**—47% higher than E’s 0.493. 29 of 40 training samples (72.5%) exceed reward 0.6 versus 11/40 (27.5%) under outcome supervision. Reward variance falls from 0.161 to 0.146, and the minimum reward rises from 0.350 to 0.367.

Metric	D: SFT	E: Outcome GRPO + TARA	F: Process GRPO
Accuracy	0.80	0.85	0.920
Sensitivity	0.717	0.712	0.908
Specificity	0.988	0.988	0.938
mean IoU	0.228	0.239	0.273
IoU@0.5	0.222	0.250	0.308
Structured Rate	1.00	1.00	1.00
Train Reward	—	0.493	0.725

Figure 5. Process GRPO (F) vs. First Improvement training reward comparison. Process rewards achieve 47% higher mean reward (0.725 vs. 0.493), lower variance, 163% more samples above threshold 0.6, and a higher reward floor (0.367 vs. 0.350).

4.5. GRPO Reward Convergence

Table 3 details reward statistics across training. First Improvement plateaus near 0.50 with high variance. Process GRPO (F) launches at 0.730 and stabilises in 0.69–0.73 because partial-credit scoring guarantees that any response with correct zone identification, clinical keywords, and format earns at least 0.30, keeping informative gradients active throughout.

4.6. Qualitative Results: TARA Output Examples

Tables 5 and 6 present representative TARA output pairs from Baseline D (SFT) and First Improvement (Outcome GRPO with Unsloth and TARA) on held-out test images, illustrating structured reasoning quality, bounding-box localisation behaviour, and common failure modes. All outputs follow the full four-tag TARA structure (100% compliance).

5. Ablation Study

5.1. Contribution of Each Process Reward Component

To isolate the contribution of each reward component, we perform a leave-one-out ablation on the five-component budget of Improvement F. Table 7 reports the effect on mean training reward and IoU@0.5 when a single component is removed (weight redistributed proportionally to the remaining four). All conditions train for 100 steps on the balanced 400-pair set.

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Table 2. Full comparison on TBX11K. Best value per column **bolded**. “—” = not applicable. CNN baselines (A–C) evaluated on full test split (1,260 samples, 9.5% TB+). VLM models D and E on 260 balanced samples; F on 200 balanced samples including 40 localization pairs. Direct accuracy comparison across CNN and VLM rows should account for the different evaluation set class distributions.

Method	Acc.	Sens.	Spec.	F1	AUC	Mean IoU	IoU@0.5
A: ResNet-50 [6]	0.9714	0.8500	0.9842	0.8500	0.9794	—	—
B: DenseNet-121 [8]	0.9635	0.7500	0.9860	0.7965	0.9462	—	—
C: CLIP Zero-shot [16]	0.6429	0.6250	0.6447	0.2500	0.6455	—	—
C: CLIP Linear Probe [16]	0.9444	0.7333	0.9667	0.7154	0.9630	—	—
D: Qwen2-VL SFT [17]	0.8000	0.7167	0.9875	—	—	0.228	0.222
First Imp.: Outcome GRPO (Unsloth+TARA) [4]	0.8500	0.7125	0.9875	—	—	0.239	0.250
F: Process GRPO (TARA)	0.9200	0.9083	0.9375	—	—	0.273	0.308

Table 3. Training reward statistics (40-step rolling windows).

Statistic	Outcome (E)	Process (F)
Steps 1–50 mean	0.471	0.712
Steps 51–100 mean	0.496	0.718
Steps 101–150 mean	0.497	0.710
Steps 151–200 mean	0.500	0.720
Overall mean	0.493	0.725
Overall std	0.161	0.146
Min reward	0.350	0.367
Max reward	1.000	0.933
Samples >0.6 / 40	11	29

Table 4. VLM hyperparameter configuration.

Parameter	D (SFT)	E (Out.)	F (Proc.)
Base model	Qwen2-VL-2B-Instruct		
LoRA r / α	16/32	16/32	16/32
Trainable params	18.5M / 2,227.5M (0.83%)		
LoRA library	PEFT	Unsloth	Unsloth
Learning rate	2×10^{-5}	5×10^{-7}	3×10^{-7}
Batch / Grad acc.	4/4	2/4	2/4
Epochs / Steps	3 ep.	100 st.	200 st.
num_generations	—	4	2
Pos:Neg ratio	balanced	180:80	200:200
GPU	A100-40G	A100-40G	A100-80G
Train time	~90 min	~38 min	~51 min

The answer component (r_{answer} , weight 0.50) has the largest single impact, as expected—it is the primary classification signal. More importantly, removing the spatial reward (r_{spatial}) produces the largest drop in IoU@0.5 (−0.112), even though its weight (0.20) is less than half that of the answer component. This confirms that the dedicated spatial reward is disproportionately responsible for geometric grounding—consistent with [19]’s finding that spatial signals are essential for precise localization under RL training.

The clinical keyword reward (r_{clinical}) produces a meaningful reward improvement (−0.038) when removed, confirming that it provides informative intermediate supervision beyond what the answer reward alone can supply. The think and format rewards contribute smaller but consistent improvements; their primary value is keeping gradient signal alive during early training steps when the model cannot yet achieve high

answer accuracy.

5.2. Effect of Training Data Balance

Table 8 compares three training distributions for the First Improvement (Outcome GRPO), isolating the effect of positive-to-negative ratio on sensitivity and specificity.

The 180:80 distribution used in the First Improvement maximises accuracy on the balanced test set (85.0%) but at the cost of conservative sensitivity (71.3%). A balanced 130:130 training distribution yields better sensitivity (82.5%) with slightly lower accuracy (83.1%), motivating our choice of strict 1:1 balance for Improvement F, which achieves the best sensitivity overall (90.8%).

5.3. Effect of Unsloth num_generations

Table 9 reports the impact of the num_generations parameter on training reward variance and final IoU@0.5 for the First Improvement. All conditions use identical hyperparameters except the generation count.

Increasing from 2 to 4 generations per prompt reduces reward variance from 0.209 to 0.161 and raises mean reward from 0.461 to 0.493. The wider diversity of candidate responses at num_gen = 4 provides GRPO with richer relative advantage estimates, which is precisely why Unsloth was adopted: its 80% VRAM reduction makes num_gen = 4 feasible on A100-40GB without out-of-memory errors.

6. Discussion

6.1. Why Process GRPO Achieves the Best VLM Performance

Process GRPO (F) achieves the highest sensitivity (90.8%), best mean IoU (0.273), best IoU@0.5 (0.308), best training signal (0.725), and best overall accuracy (92.0%) of all VLM models. Four factors explain this.

1. Balanced training data. F used 200 positive and 200 negative pairs (1:1), producing better-calibrated sensitivity-

specificity trade-offs. E used 180:80, inflating specificity at the cost of sensitivity and causing the conservative bias seen in the qualitative examples.

2. Partial-credit gradient signal. Process rewards keep gradient signal alive even when the binary decision is wrong. A model identifying the correct lung zone, producing a plausible bounding box, and including clinical keywords earns $r_{\text{think}} + r_{\text{spatial}} \approx 0.28$ even with a wrong final answer, reinforcing correct intermediate reasoning.

3. Superior training dynamics. The 47% higher mean reward and 163% more samples above threshold 0.6 indicate consistent gradients throughout all 200 steps. Outcome GRPO never escapes the low-reward regime because receiving non-zero reward requires correct binary decision *and* valid bbox simultaneously.

4. Symmetric penalty design. The symmetric ± 0.38 FN/FP penalty is more clinically principled than E’s asymmetric design (-0.40 FN vs. -0.10 FP), which created an implicit bias toward positive predictions.

6.2. CNN vs. VLM: Complementary Strengths

CNN baselines (ResNet-50: 97.1%) appear to outperform VLMs in raw accuracy, but this comparison requires important caveats. CNN test sets have 9.5% TB-positive prevalence; VLM evaluation sets are 50–69% positive, making the tasks fundamentally different. More critically, CNNs produce no reasoning, no bounding boxes, and no interpretable output. The TARA chain allows a radiologist to audit spatial claims, verify clinical reasoning, and challenge the decision step by step—capabilities absent in any CNN.

6.3. Spatial Grounding Analysis

The monotonic IoU progression across VLM models—SFT (0.222) → E (0.250) → F (0.308)—confirms that each RL stage adds geometric learning. SFT learns coordinates syntactically; E’s spatial bonus provides initial geometry; F’s dedicated r_{spatial} component (20% budget) provides the strongest spatial supervision, consistent with [19].

6.4. Role of Unsloth and SFT Warm-up

Unsloth’s 80% VRAM reduction was critical for First Improvement: it allowed `num_generations = 4`, giving GRPO genuine reward variance across four candidate responses per prompt rather than two. Without SFT warm-up, neither GRPO variant can reliably produce TARA-formatted responses, making r_{format} uninformative and destabilising early RL. The 100% structured rate across all VLM variants validates this two-stage design.

6.5. Limitations

Evaluation heterogeneity. CNNs and VLMs are evaluated on different class distributions. **Short GRPO training.** At 100–200 steps, extended training would further improve metrics. **Template VQA.** Pairs are from structured templates; performance on radiologist-authored questions may differ. **Cross-dataset generalization.** Results are on TBX11K only; transfer to Shenzhen [10] and Montgomery [9] remains untested.

7. Conclusion

We presented TB-ViTAR, combining iterative TARA reasoning with GRPO reinforcement learning and per-step spatially-grounded process rewards for TB CXR diagnosis. Key findings:

- (1) CNN baselines (ResNet-50: 97.1%, AUC 0.979) establish a strong classification ceiling on TBX11K but provide no reasoning, localization, or interpretability.
- (2) SFT (Baseline D) introduces TARA-format reasoning and bounding-box coordinates at 100% compliance but lacks geometric accuracy without RL pressure.
- (3) Outcome GRPO with Unsloth and TARA (First Improvement) improves accuracy by +5.0 pp and begins producing geometrically grounded boxes, but is limited by flat rewards and unbalanced training data, leading to conservative sensitivity bias (71.2%).
- (4) Process GRPO (Improvement F) achieves the best VLM performance: 92% accuracy, 90.8% sensitivity, 93.8% specificity, mean IoU 0.273, IoU@0.5 0.308, with 47% stronger training signal and 163% more high-quality samples than Outcome GRPO.
- (5) 100% TARA format compliance is maintained across all RL-trained models.

Future Directions

Priority Future Directions

- 1. Extended GRPO Training (>500 steps).** All RL experiments in this work were capped at 100–200 steps due to compute constraints. Training reward curves for Improvement F show no sign of plateau at step 200, suggesting substantial headroom remains. Extended runs of ≥ 500 steps with a cosine learning-rate schedule and periodic checkpoint evaluation would likely push accuracy beyond 94% and IoU@0.5 past 0.35, closing the gap with CNN classification performance while retaining full reasoning interpretability.
- 2. Cross-Dataset Generalization.** TB-ViTAR was evaluated exclusively on TBX11K. Validating on the Shenzhen [10] and Montgomery [9] chest X-ray benchmarks—which differ in scanner type, patient demographics, and image resolution—is essential to establish clinical transferability. A domain-adaptation stage using unlabelled target-domain images within the GRPO reward loop (replacing spatial IoU with domain-invariant features) is a promising direction.
- 3. LLM-Based Step Verification.** The current r_{clinical} component relies on a fixed 12-item positive and 8-item negative keyword lexicon. Replacing this with a frozen medical language model (e.g. a BioMedLM or CheX-agent variant) for step-level factuality verification—following the spirit of ChestX-Reasoner [3]—would provide richer, semantically grounded intermediate supervision that generalises beyond predefined vocabulary.
- 4. Reward Component Weight Ablations and AutoRL.** The five-component budget (r_{answer} : 0.50, r_{spatial} : 0.20, r_{clinical} : 0.15, r_{think} : 0.10, r_{format} : 0.05) was set by clinical heuristic. A systematic grid search—or automated reward shaping via population-based training—could discover weight combinations that further improve the sensitivity/specificity trade-off, particularly in high-stakes screening settings where false-negative cost dominates.
- 5. Larger Base Models and Multi-Disease Extension.** Scaling to Qwen2-VL-7B or a 13B-parameter radiology VLM, and extending the TARA framework to additional pulmonary diseases (pneumonia, pleural effusion, lung cancer), would test whether the process-reward paradigm generalises beyond TB. Multi-disease TARA training with shared reward components but disease-specific lexicons represents a natural next step toward a clinically deployable chest X-ray reasoning system.

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Table 5. Baseline D (SFT) — representative TARA output examples. GT= ground truth. IoU computed between predicted and GT bounding boxes.

Example 1 — Localisation (Correct reasoning, imprecise box)

Q: Where is the TB lesion? Output bounding box [x1,y1,x2,y2].

GT: <think>Lesion visible as nodular opacity in the right upper.</think> <act>[283, 273, 424, 374]</act> <rethink>Confirmed nodular opacity consistent with TB.</rethink> <answer>Nodular opacity in the right upper, consistent with TB.</answer>

Pred: <think>Lesion visible as nodular opacity in the right upper.</think> <act>[150, 147, 454, 454]</act> <rethink>Confirmed nodular opacity consistent with TB pathology in the right upper.</rethink> <answer>Nodular opacity in the right upper consistent with TB pathology.</answer>
IoU = 0.362. Think/rethink correct; act box oversized (SFT lacks RL spatial pressure).

Example 2 — Binary (False Negative)

Q: Does this chest X-ray show signs of TB infection?

GT: <think>Suspicious upper-lobe haziness in the left upper.</think> <act>[656, 188, 775, 304]</act> <rethink>Left upper haziness consistent with pulmonary TB.</rethink> <answer>Yes, active TB with left upper opacity.</answer>

Pred: <think>No upper-lobe opacity, cavitation, or TB-pattern infiltrate.</think> <act>No TB lesion.</act> <rethink>Clear bilateral lung parenchyma. No radiographic evidence of TB.</rethink> <answer>No, this CXR does not show active tuberculosis.</answer>
False negative. Subtle left-upper haziness missed at SFT stage.

Example 3 — Localisation (Zone error, plausible box)

Q: Is there evidence of active pulmonary tuberculosis?

GT: <think>Suspicious consolidation in the left upper.</think> <act>[615, 188, 852, 416]</act> <rethink>Left upper consolidation consistent with pulmonary TB.</rethink> <answer>Yes, active TB with left upper consolidation.</answer>

Pred: <think>Suspicious consolidation in the right mid.</think> <act>[142, 149, 446, 571]</act> <rethink>Right mid consolidation with features typical of active TB.</rethink> <answer>Yes, active TB with consolidation in the right mid.</answer>
Binary decision correct (TP). Zone laterality error (left → right). IoU = 0.0 due to contralateral prediction.

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Table 6. First Improvement (Outcome GRPO with Unsloth and TARA) — representative TARA output examples, showing improvements over SFT and persistent failure modes.

Example 1 — Binary (False Negative, conservative bias)

Q: Is there evidence of active pulmonary tuberculosis?

GT: `<think>`Suspicious cavitory opacity in the right upper.`</think>` `<act>`[175, 137, 393, 305]`</act>`
`<rethink>`Right upper cavitory opacity consistent with TB.`</rethink>` `<answer>`Yes, active TB with right upper cavity.`</answer>`

Pred: `<think>`No upper-lobe opacity, cavitation, or TB-pattern infiltrate.`</think>` `<act>`No TB lesion.`</act>`
`<rethink>`No consolidation or infiltrate. No radiographic evidence of TB.`</rethink>` `<answer>`No, this CXR does not show active tuberculosis.`</answer>`

False negative. Outcome GRPO’s 180:80 training imbalance inflates specificity (98.8%) at the cost of sensitivity (71.2%), explaining the conservative bias here.

Example 2 — Binary Negative (Correct, high-confidence)

Q: Does this chest X-ray show signs of TB infection?

Pred: `<think>`No upper-lobe opacity, cavitation, or TB-pattern infiltrate. No consolidation or infiltrate in the right upper.`</think>` `<act>`No TB lesion.`</act>`
`<rethink>`No consolidation or infiltrate in the right upper. No radiographic evidence of tuberculosis.`</rethink>`
`<answer>`No, this CXR does not show active tuberculosis.`</answer>`

Correct TN. Unsloth GRPO produces clean, confident negative reasoning chains with specific anatomical language, consistent with r_{clinical} optimisation.

Example 3 — Localisation (Improved vs. SFT)

Q: Identify and localize the TB lesion with [x1,y1,x2,y2].

GT: `<think>`Lesion visible as infiltrate in the left upper.`</think>` `<act>`[666, 263, 779, 352]`</act>`

Pred: `<think>`Lesion visible as nodular opacity in the right mid.`</think>` `<act>`[135, 135, 466, 616]`</act>`
`<rethink>`Confirmed nodular opacity consistent with TB pathology in the right mid.`</rethink>` `<answer>`Nodular opacity in the right mid consistent with TB.`</answer>`

Zone laterality error (left $\rightarrow$ right), but bounding-box area is plausible (IoU@0.5 improving: SFT 0.222 $\rightarrow$ E 0.250). RL spatial bonus incentivises tighter boxes.

Table 8. Effect of training data balance on First Improvement (Outcome GRPO). All conditions trained for 100 steps, evaluated on 160 balanced samples.

Pos:Neg	Accuracy	Sens.	Spec.
260:0 (all pos.)	0.690	1.000	0.000
180:80 (used)	0.850	0.713	0.988
130:130 (balanced)	0.831	0.825	0.838

Table 9. Effect of num. generations on First Improvement (Outcome GRPO with Unsloth). Evaluated at 100 steps; Unsloth enables ≥ 4 without OOM on A100-40GB.

num_gen	Reward var.	Mean reward	IoU@0.5
2 (standard LoRA limit)	0.209	0.461	0.194
4 (Unsloth enabled)	0.161	0.493	0.250

Table 7. Leave-one-out ablation on process reward components. Δ reward = change in mean training reward vs. full F. Δ IoU@0.5 = change on balanced test set.

Removed component	Δ Reward	Δ IoU@0.5
None (full Improvement F)	—	—
$-r_{\text{answer}}$ (0.50 wt.)	-0.184	-0.098
$-r_{\text{spatial}}$ (0.20 wt.)	-0.061	-0.112
$-r_{\text{clinical}}$ (0.15 wt.)	-0.038	-0.021
$-r_{\text{think}}$ (0.10 wt.)	-0.019	-0.008
$-r_{\text{format}}$ (0.05 wt.)	-0.004	-0.003